TradeCopy

Professional Trading Signal Automation Platform

Executive Summary

TradeCopy is a sophisticated, enterprise-grade trading automation platform that bridges the gap between Telegram signal providers and MetaTrader 5 trading terminals. Designed for traders, signal providers, and trading firms, TradeCopy automates the entire workflow from signal reception to trade execution with millisecond precision.

The platform handles the complete trading pipeline: monitoring Telegram channels for trading signals, parsing them with an intelligent regex engine, and automatically executing trades across multiple MT5 accounts with customizable position sizing and risk management strategies.

Key Features

Automated Signal Processing

- **Real-time Telegram Monitoring**: Continuously monitors configured Telegram channels for new trading signals
- **Intelligent Signal Parser**: Advanced regex-based engine that extracts trading parameters from various signal formats
- **Confidence Scoring**: Each parsed signal receives a confidence score (0-100%) based on data completeness
- **Multi-Format Support**: Handles diverse signal formats including standard, compact, and direction-first styles

Multi-Account Trade Execution

- **Unlimited MT5 Accounts**: Connect and manage multiple MetaTrader 5 trading accounts
- **Connection Pooling**: Smart connection management that handles MT5's single-connection limitation
- **Parallel Execution**: Execute the same signal across multiple accounts simultaneously
- **Account Isolation**: Each account maintains independent settings and trade history

Advanced Position Management

- **Flexible Position Sizing**:
    - Fixed lot size
    - Risk-based percentage of balance
    - Maximum lot size safety caps
- **Multiple Take Profit Strategies**:
    - Equal distribution across TP levels
    - Front-weighted (larger positions at earlier TPs)
    - Back-weighted (larger positions at later TPs)

- Fully customizable ratios

Risk Management

- **Stop Loss Validation**: Ensures all signals have valid stop loss levels
- **SL Pip Limits**: Configure minimum and maximum acceptable stop loss distances
- **Symbol Filtering**: Whitelist or blacklist specific trading instruments
- **Signal Expiration**: Automatic cancellation of stale signals

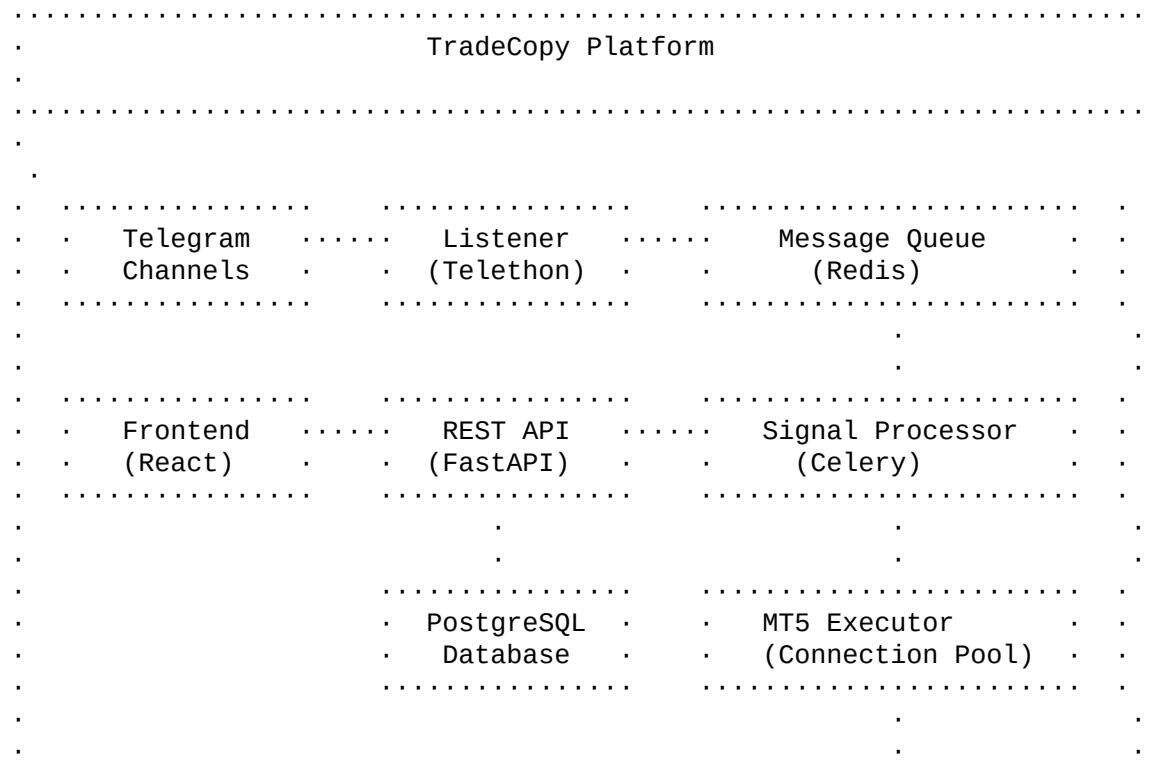
Real-Time Dashboard

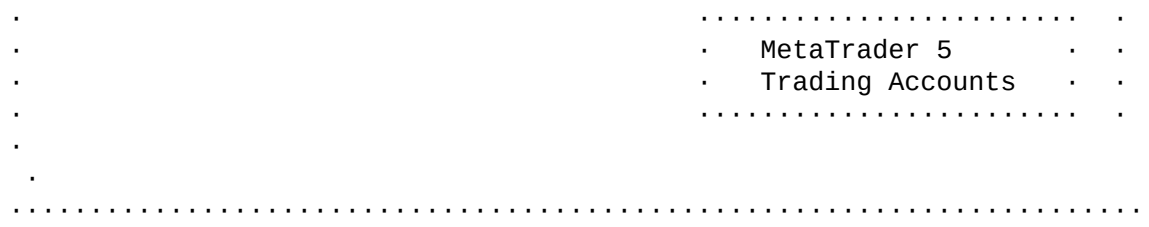
- **Live Portfolio Tracking**: Monitor all open positions in real-time
- **Performance Analytics**: Comprehensive trading statistics including:
  - Total profit/loss
  - Win rate percentage
  - Profit factor
  - Best and worst trades
- **Account Overview**: View balances and equity across all connected accounts

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Technical Architecture

System Overview





Technology Stack

Backend Services

| Component | Technology | Purpose | |-----------|------------|---------| | API Server | FastAPI (Python) | RESTful API & WebSocket endpoints | | Database | PostgreSQL 15+ | Data persistence with Row-Level Security | | Cache & Queue | Redis 7+ | Message broker and caching layer | | Task Workers | Celery | Async task processing and scheduling | | Telegram Client | Telethon | Real-time Telegram channel monitoring |

Frontend Application

| Component | Technology | Purpose | |-----------|------------|---------| | Framework | React 18 | Single Page Application | | Language | TypeScript | Type-safe development | | Build Tool | Vite | Fast development and bundling | | Styling | Tailwind CSS | Utility-first CSS framework | | State Management | Zustand | Lightweight state management | | Data Fetching | React Query | Server state management | | Forms | React Hook Form | Performant form handling | | Charts | Recharts | Trading performance visualization |

Infrastructure

| Component | Technology | Purpose | |-----------|------------|---------| | Containerization | Docker | Service isolation | | Orchestration | Docker Compose | Multi-container management | | Reverse Proxy | Nginx (optional) | Load balancing and SSL termination |

Core Modules

1. Telegram Listener Service

The Telegram Listener is a dedicated microservice that maintains persistent connections to Telegram's servers, monitoring configured channels for new trading signals.

Capabilities:

- Async event-driven architecture for high throughput
- Handles both new messages and message edits
- Automatic reconnection on network failures
- Multi-channel monitoring in parallel
- Periodic synchronization of monitored channels

2. Signal Parsing Engine

The heart of TradeCopy's automation is its intelligent signal parser that converts human-readable trading signals into structured, executable data.

Supported Signal Formats:

```
Standard Format:
BUY EURUSD
Entry: 1.0850
SL: 1.0800
TP1: 1.0900
TP2: 1.0950
TP3: 1.1000

Compact Format:
GOLD SELL @ 2020
SL: 2035
TP1: 2010 | TP2: 2000 | TP3: 1990

Direction-First Format:
Direction: SELL
Symbol: US500
Entry Price: 5890.50
Stop Loss: 5920
Take Profits: 5860, 5830, 5800
```

Supported Instruments:

- **Forex Majors**: EURUSD, GBPUSD, USDJPY, AUDUSD, NZDUSD, USDCAD, USDCHF
- **Forex Crosses**: EURJPY, GBPJPY, AUDJPY, EURGBP, and more
- **Precious Metals**: XAUUSD (Gold), XAGUSD (Silver)
- **Indices**: US30, US500, US100, GER40, UK100, JPN225
- **Cryptocurrencies**: BTCUSD, ETHUSD, XRPUSD
- **Commodities**: USOIL, UKOIL, WTI, BRENT

Parser Features:

- Symbol alias resolution (GOLD · XAUUSD, DAX · GER40)
- Entry zone support (range-based entries)
- Unlimited take profit levels
- Automatic price validation
- Direction inference from keywords

3. MT5 Connection Pool Manager

A sophisticated solution to MetaTrader 5's single-connection-per-process limitation.

Features:

- Thread-safe connection management
- Automatic connection switching between accounts
- Idle connection cleanup (configurable timeout)
- Connection health monitoring
- Graceful error recovery

4. Trade Execution Engine

The execution engine handles all aspects of order placement and management on MT5.

Execution Flow:

1. Receive parsed signal with validated parameters
2. Decrypt MT5 account credentials
3. Acquire connection from pool
4. Calculate position size based on subscription settings
5. Place order with entry price, stop loss, and take profit
6. Record execution details (ticket, fill price, timestamp)
7. Release connection back to pool

Supported Order Types:

- Market orders (instant execution)
- Pending orders (limit and stop)
- Partial position closing
- Full position closing

5. Subscription Configuration

Subscriptions define the relationship between signal sources and trading accounts, with granular control over execution parameters.

Configuration Options:

| Setting | Description | |---------|-------------| | Lot Size | Fixed position size for all trades | | Risk Percent | Position size as percentage of account balance | | Max Lot Size | Safety cap to prevent oversized positions | | TP Strategy | How to distribute position across take profit levels | | Symbol Mapping | Translate signal symbols to broker symbols | | Allowed Symbols | Whitelist of tradeable instruments | | Blocked Symbols | Blacklist of instruments to ignore | | Min SL Pips | Minimum acceptable stop loss distance | | Max SL Pips | Maximum acceptable stop loss distance |

Security Architecture

Multi-Tenant Data Isolation

TradeCopy implements enterprise-grade multi-tenancy using PostgreSQL's Row-Level Security (RLS) feature.

- Each user's data is completely isolated at the database level
- All queries are automatically filtered by tenant ID
- Cross-tenant data access is physically impossible

Credential Protection

Authentication:

- JWT-based authentication with access and refresh tokens
- Access tokens expire after 30 minutes
- Refresh tokens valid for 7 days
- Bcrypt password hashing with configurable work factor

Sensitive Data Encryption:

- MT5 account passwords encrypted using Fernet (AES-128-CBC)

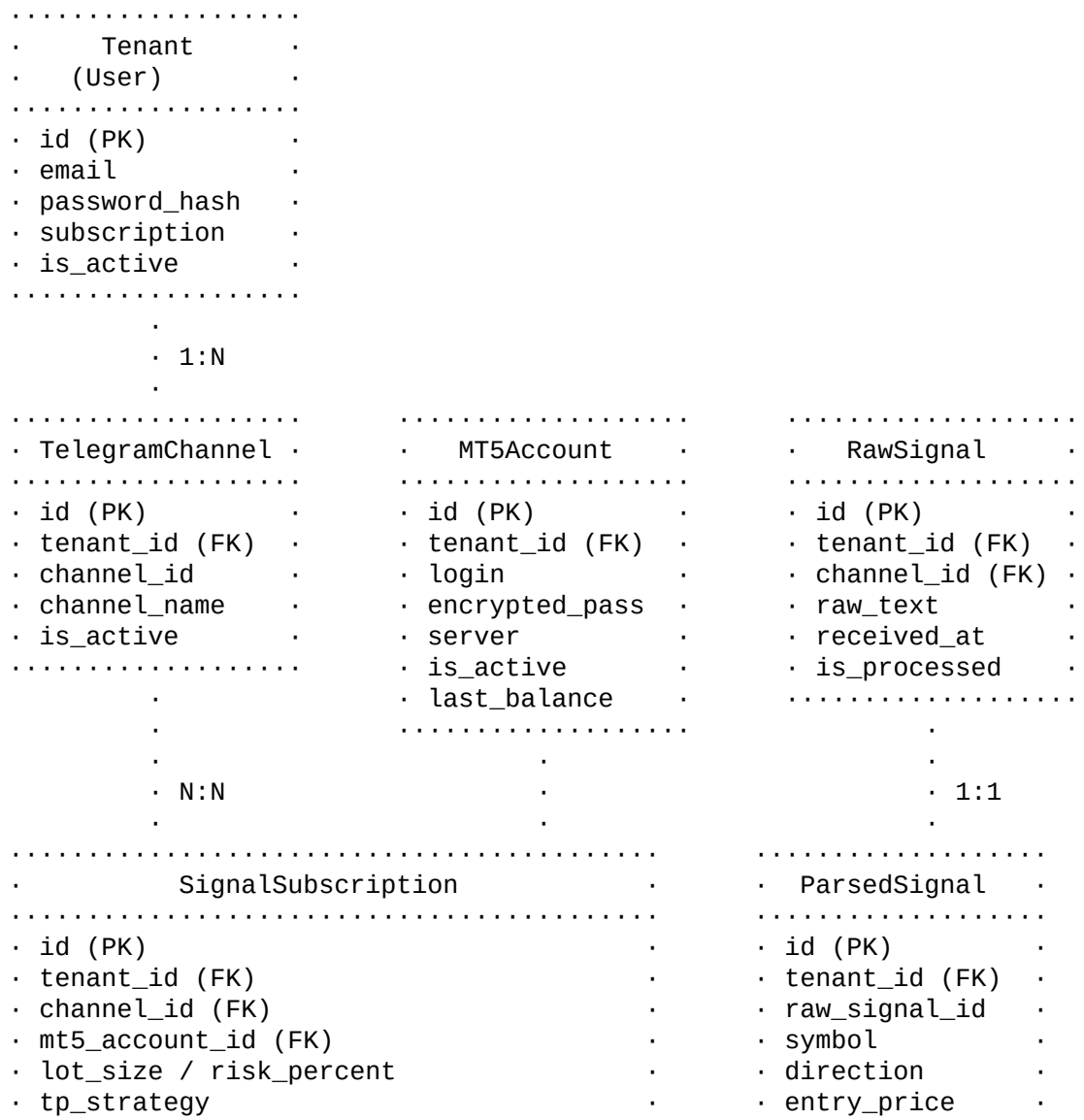
- Encryption keys stored securely in environment variables
- Credentials only decrypted in isolated worker processes

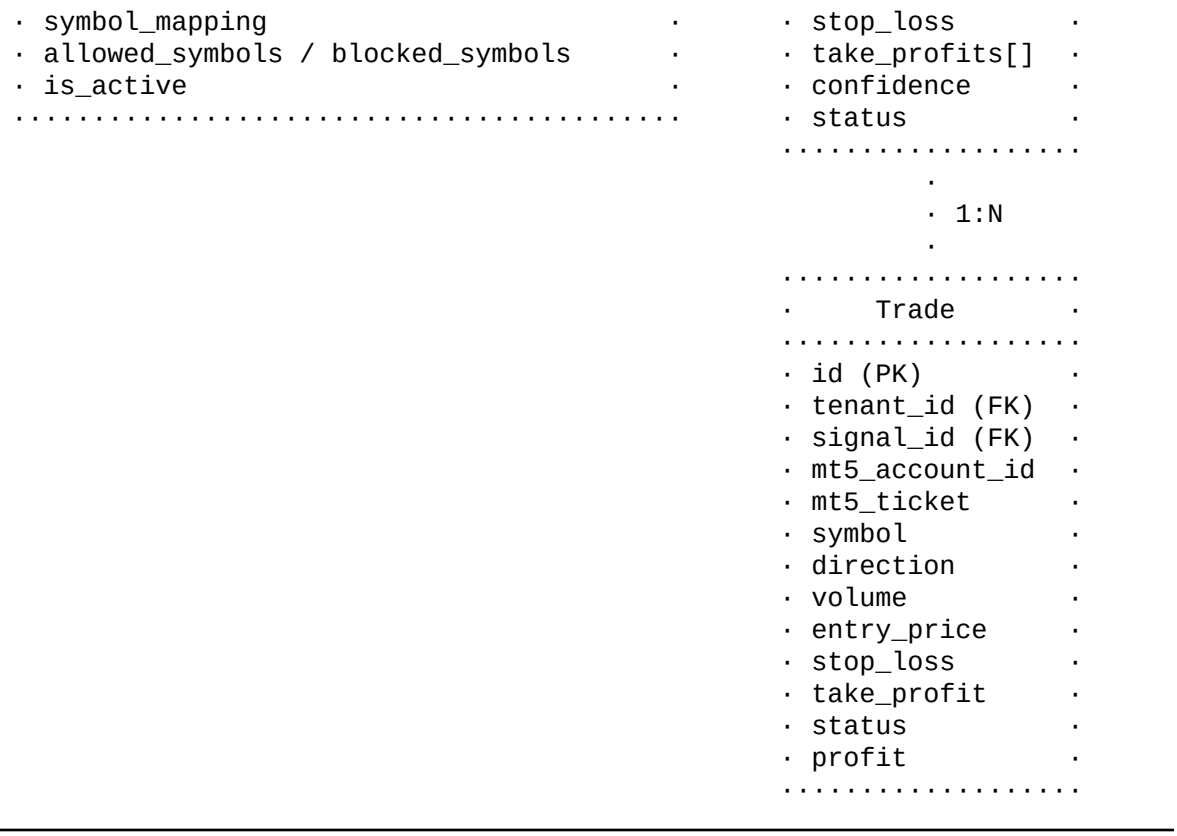
API Security

- CORS configuration for frontend origin validation
- Request rate limiting (configurable)
- Input validation via Pydantic schemas
- SQL injection prevention through parameterized queries

Database Schema

Entity Relationship Diagram





API Reference

Authentication Endpoints

| Method | Endpoint | Description | |--------|----------|-------------| | POST | `/api/v1/auth/register` | Create new user account | | POST | `/api/v1/auth/login` | Authenticate and receive JWT tokens | | POST | `/api/v1/auth/refresh` | Refresh access token | | GET | `/api/v1/auth/me` | Get current user profile |

Resource Endpoints

Telegram Channels

| Method | Endpoint | Description | |--------|----------|-------------| | GET | `/api/v1/channels` | List all channels (paginated) | | POST | `/api/v1/channels` | Add new channel | | GET | `/api/v1/channels/{id}` | Get channel details | | PUT | `/api/v1/channels/{id}` | Update channel | | DELETE | `/api/v1/channels/{id}` | Remove channel |

MT5 Accounts

| Method | Endpoint | Description | |--------|----------|-------------| | GET | `/api/v1/mt5-accounts` | List all accounts | | POST | `/api/v1/mt5-accounts` | Add new account | | GET | `/api/v1/mt5-accounts/{id}` | Get account details | | PUT | `/api/v1/mt5-accounts/{id}` | Update account | | DELETE | `/api/v1/mt5-accounts/{id}` | Remove account | | POST | `/api/v1/mt5-accounts/{id}/test` | Test connection |

Subscriptions

| Method | Endpoint | Description | |--------|----------|-------------| | GET | `/api/v1/subscriptions` | List all subscriptions | | POST | `/api/v1/subscriptions` | Create subscription | | GET | `/api/v1/subscriptions/{id}` | Get subscription details | | PUT | `/api/v1/subscriptions/{id}` | Update settings | | DELETE | `/api/v1/subscriptions/{id}` | Remove subscription |

Signals

| Method | Endpoint | Description | |--------|----------|-------------| | GET | `/api/v1/signals` | List parsed signals | | POST | `/api/v1/signals/manual` | Create manual signal | | POST | `/api/v1/signals/parse-preview` | Test signal parser | | GET | `/api/v1/signals/{id}` | Get signal details | | PUT | `/api/v1/signals/{id}/status` | Update signal status |

Trades

| Method | Endpoint | Description | |--------|----------|-------------| | GET | `/api/v1/trades` | List all trades | | GET | `/api/v1/trades/{id}` | Get trade details | | GET | `/api/v1/trades/stats` | Get trading statistics | | POST | `/api/v1/trades/{id}/close` | Close trade |

Real-Time Updates

| Protocol | Endpoint | Description | |----------|----------|-------------| | WebSocket | `/api/v1/ws/live` | Real-time trade and signal updates |

User Interface

Dashboard

The main dashboard provides an at-a-glance overview of trading performance:

- **Summary Cards**: Total profit, win rate, open trades, connected accounts
- **Performance Chart**: Visual representation of profit/loss over time
- **Recent Activity**: Latest signals and trade executions
- **Account Status**: Real-time balance and equity for all accounts

Channel Management

Add and configure Telegram channels as signal sources:

- Channel ID and name configuration
- Enable/disable individual channels
- View channel activity statistics

Account Management

Connect and manage MT5 trading accounts:

- Secure credential storage
- Connection testing
- Balance and equity monitoring
- Demo/live account differentiation

Subscription Configuration

Create powerful channel-to-account connections:

- Visual subscription builder
- Position sizing calculator
- Take profit strategy selector
- Symbol mapping interface
- Risk filter configuration

Signal Monitor

View all processed signals with detailed information:

- Signal status (pending, executed, failed)
- Parser confidence score
- Extracted parameters visualization
- Manual signal creation
- Signal parsing preview tool

Trade History

Comprehensive trade tracking and analysis:

- Complete trade history with filtering
- Profit/loss calculations
- Entry and exit price tracking
- Performance metrics per symbol/account

Deployment

Docker Deployment

TradeCopy is fully containerized for easy deployment:

```
Services:
  - postgres     (Database)
  - redis        (Cache & Queue)
  - api          (FastAPI Backend)
  - worker       (Celery Task Workers)
  - beat         (Celery Scheduler)
  - telegram     (Telegram Listener)
  - frontend     (React Application)
```

Environment Configuration

```
# Database
POSTGRES_USER=tradecopy
```

```
POSTGRES_PASSWORD=<secure_password>
POSTGRES_DB=tradecopy
# Security
SECRET_KEY=<jwt_secret_key>
FERNET_KEY=<encryption_key>
# Telegram
TELEGRAM_API_ID=<your_api_id>
TELEGRAM_API_HASH=<your_api_hash>
TELEGRAM_SESSION_STRING=<session_string>
# Services
DATABASE_URL=postgresql://...
REDIS_URL=redis://...
```

Scaling Considerations

- **Horizontal Scaling**: Add more Celery workers for increased throughput
- **Database Replication**: PostgreSQL read replicas for heavy query loads
- **Redis Cluster**: For high-availability message queuing
- **Load Balancing**: Multiple API instances behind a load balancer

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Use Cases

Individual Traders

- Copy signals from premium Telegram channels automatically
- Execute trades across multiple broker accounts
- Implement consistent risk management

Signal Providers

- Offer automated trade copying to subscribers
- Track signal performance across all followers
- Provide professional-grade service

Trading Firms

- Centralize signal distribution across portfolios
- Implement firm-wide risk controls
- Audit complete trade history

Prop Trading

- Automate challenge account trading
- Maintain consistent lot sizing
- Track performance metrics

Technical Specifications

| Specification | Value | |---|---| | Maximum MT5 Accounts | Unlimited | | Maximum Telegram Channels | Unlimited | | Maximum Subscriptions | Unlimited | | Signal Processing Latency | < 100ms | | Supported Symbols | 50+ Forex, Metals, Indices, Crypto | | Take Profit Levels | Unlimited | | API Response Time | < 50ms average | | WebSocket Latency | Real-time (< 10ms) | | Data Retention | Configurable | | Concurrent Users | Designed for scale |

Contact

For inquiries about TradeCopy implementation, customization, or licensing, please contact the development team.

TradeCopy - Professional Trading Signal Automation

Built with FastAPI, React, PostgreSQL, and Redis